

Part 2: Conflict Analysis and Proofs

ACP Summer School 2026

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Last time...

Constraint Programming

Propagators and Explanations

Linear inequality: $\sum w_i x_i \geq c$

Max: $x = \max(y, z)$

All-Different

Checkers

Propagation based on Hall sets!

$$x_1 \in \{1, 2\}$$

$$x_2 \in \{1, 2, \cancel{3}\}$$

$$\langle x_3 \geq 3 \rangle \wedge \langle x_3 \leq 5 \rangle \wedge \\ \langle x_4 \geq 3 \rangle \wedge \langle x_4 \leq 5 \rangle \wedge \\ \langle x_5 \geq 3 \rangle \wedge \langle x_3 \leq 5 \rangle$$

$\Rightarrow$

$$\langle x_2 \neq 3 \rangle$$

$$x_3 \in \{3, 4\}$$

$$x_4 \in \{3, 4, 5\}$$

$$x_5 \in \{4, 5\}$$

Checker for All-Different

1. Reconstruct domains based on the explanation

2. Check condition

$$H = \{x_i\}$$

$$\bigcup D(x_i) < |H| ?$$

In this lecture...

Conflict Analysis —————→ **Proof-theoretic view of CP**

**Derive new constraints
by combining constraints**

Certificates

Trail

$$\langle x_9 \leq 0 \rangle \quad @1$$

$$\langle x_1 \leq 2 \rangle \quad @2$$

$$\langle x_2 \leq 3 \rangle \quad @3$$

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle y \geq 2 \rangle \quad @4$$

$$\langle x_5 \leq 2 \rangle \quad @5$$

$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

Constraints

$$c_1 : 3x_2 + x_3 + x_9 \geq 10$$

$$c_2 : -x_3 + 5x_4 \geq 4$$

$$c_3 : x_1 + x_5 + 2x_6 \geq 8$$

$$c_4 : x_7 - x_6 \geq 0$$

$$c_5 : 2x_8 - x_6 + x_9 \geq 0$$

$$c_6 : -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$x_i, y \in \{-10, 10\}$$

Trail

$$\langle x_9 \leq 0 \rangle \quad @1$$

$$\langle x_1 \leq 2 \rangle \quad @2$$

$$\langle x_2 \leq 3 \rangle \quad @3$$

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle y \geq 2 \rangle \quad @4$$

$$\langle x_5 \leq 2 \rangle \quad @5$$

$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

info
about
search

Constraints

$$c_1 : 3x_2 + x_3 + x_9 \geq 10$$

$$c_2 : -x_3 + 5x_4 \geq 4$$

$$c_3 : x_1 + x_5 + 2x_6 \geq 8$$

$$c_4 : x_7 - x_6 \geq 0$$

$$c_5 : 2x_8 - x_6 + x_9 \geq 0$$

$$c_6 : -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$x_i, y \in \{-10, 10\}$$

Trail

$$\langle x_9 \leq 0 \rangle$$

$$\langle x_1 \leq 2 \rangle$$

$$\langle x_2 \leq 3 \rangle$$

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle y \geq 2 \rangle$$

$$\langle x_5 \leq 2 \rangle$$

$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

@1

@2

@3

@4

@5

Constraints

Decided $x_9 \leq 0$

Decision level 1

$$c_1 : 3x_2 + x_3 + x_9 \geq 10$$

$$c_2 : -x_3 + 5x_4 \geq 4$$

$$c_3 : x_1 + x_5 + 2x_6 \geq 8$$

$$c_4 : x_7 - x_6 \geq 0$$

$$c_5 : 2x_8 - x_6 + x_9 \geq 0$$

$$c_6 : -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$x_i, y \in \{-10, 10\}$$

Trail

$$\langle x_9 \leq 0 \rangle$$

$$\langle x_1 \leq 2 \rangle$$

$$\langle x_2 \leq 3 \rangle$$

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle y \geq 2 \rangle$$

$$\langle x_5 \leq 2 \rangle$$

$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

@1

@2

@3

@4

@5

Decided $x_2 \leq 3$

Decision level 3

Constraints

$$c_1 : 3x_2 + x_3 + x_9 \geq 10$$

$$c_2 : -x_3 + 5x_4 \geq 4$$

$$c_3 : x_1 + x_5 + 2x_6 \geq 8$$

$$c_4 : x_7 - x_6 \geq 0$$

$$c_5 : 2x_8 - x_6 + x_9 \geq 0$$

$$c_6 : -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$x_i, y \in \{-10, 10\}$$

Trail

$$\langle x_9 \leq 0 \rangle$$

@1

$$\langle x_1 \leq 2 \rangle$$

@2

$$\langle x_2 \leq 3 \rangle$$

@3

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

Constraint c_1

Propagated $x_3 \geq 1$

Decision level 3

$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

Constraints

$$c_1 : 3x_2 + x_3 + x_9 \geq 10$$

$$c_2 : -x_3 + 5x_4 \geq 4$$

$$c_3 : x_1 + x_5 + 2x_6 \geq 8$$

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$$c_5 : 2x_8 - x_6 + x_9 \geq 0$$

$$c_6 : -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$x_i, y \in \{-10, 10\}$$

Trail

$$\langle x_9 \leq 0 \rangle$$

@1

$$\langle x_1 \leq 2 \rangle$$

@2

$$\langle x_2 \leq 3 \rangle$$

@3

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle y \geq 2 \rangle$$

$$\langle x_5 \leq 2 \rangle$$

@5

$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

Constraints

$$c_1 : 3x_2 + x_3 + x_9 \geq 10$$

$$x_3 + 5x_4 \geq 4$$

$$+x_5 + 2x_6 \geq 8$$

$$-x_6 \geq 0$$

$$c_5 : 2x_8 - x_6 + x_9 \geq 0$$

$$c_6 : -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$x_i, y \in \{-10, 10\}$$

Constraint c_2

Propagated $x_4 \geq 1$

Decision level 3

Trail

$$\langle x_9 \leq 0 \rangle$$

@1

$$\langle x_1 \leq 2 \rangle$$

@2

$$\langle x_2 \leq 3 \rangle$$

@3

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle y \geq 2 \rangle$$

@4

$$\langle x_5 \leq 2 \rangle$$

@5

$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

**Constraint c_6
Declares conflict**

Constraints

$$c_1 : 3x_2 + x_3 + x_9 \geq 10$$

$$c_2 : -x_3 + 5x_4 \geq 4$$

$$c_3 : x_1 + x_5 + 2x_6 \geq 8$$

$$c_4 : x_7 - x_6 \geq 0$$

$$c_5 : 2x_8 - x_6 + x_9 \geq 0$$

$$c_6 : -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$x_i, y \in \{-10, 10\}$$

Trail

$$\langle x_9 \leq 0 \rangle$$

@1

$$\langle x_1 \leq 2 \rangle$$

@2

$$\langle x_2 \leq 3 \rangle$$

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle y \geq 2 \rangle$$

@4

$$\langle x_5 \leq 2 \rangle$$

@5

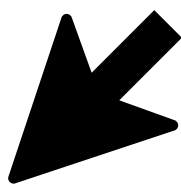
$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

Classical
Backtracking



Constraints

$$c_1 : 3x_2 + x_3 + x_9 \geq 10$$

$$c_2 : -x_3 + 5x_4 \geq 4$$

$$c_3 : x_1 + x_5 + 2x_6 \geq 8$$

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$$c_5 : 2x_8 - x_6 + x_9 \geq 0$$

$$c_6 : -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$x_i, y \in \{-10, 10\}$$

Trail

$$\langle x_9 \leq 0 \rangle$$

@1

$$\langle x_1 \leq 2 \rangle$$

@2

$$\langle x_2 \leq 3 \rangle$$

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle y \geq 2 \rangle$$

@4

~~$$\langle x_5 \leq 2 \rangle$$~~

~~@5~~

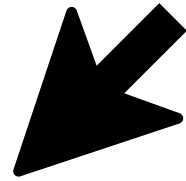
$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

Classical
Backtracking



Undo
decision

Constraints

$$c_1 : 3x_2 + x_3 + x_9 \geq 10$$

$$c_2 : -x_3 + 5x_4 \geq 4$$

$$c_3 : x_1 + x_5 + 2x_6 \geq 8$$

$$c_4 : x_7 - x_6 \geq 0$$

$$c_5 : 2x_8 - x_6 + x_9 \geq 0$$

$$c_6 : -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$x_i, y \in \{-10, 10\}$$

Trail

$$\langle x_9 \leq 0 \rangle$$

@1

$$\langle x_1 \leq 2 \rangle$$

@2

$$\langle x_2 \leq 3 \rangle$$

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle y \geq 2 \rangle$$

@4

$$\langle x_5 \geq 3 \rangle$$

@5

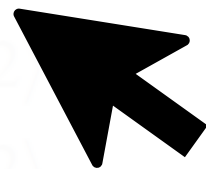
$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

Classical
Backtracking



Post negation
and continue

Constraints

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$$x_i, y \in \{-10, 10\}$$

Trail

$$\langle x_9 \leq 0 \rangle \quad @1$$

$$\langle x_1 \leq 2 \rangle \quad @2$$

$$\langle x_2 \leq 3 \rangle \quad @3$$

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle y \geq 2 \rangle \quad @4$$

$$\langle x_5 \leq 2 \rangle \quad @5$$

$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

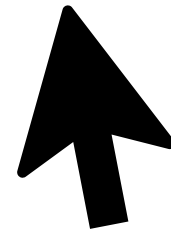
$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

Conflict Analysis

$$\overset{@3}{\langle x_3 \geq 1 \rangle} \wedge \overset{@3}{\langle x_4 \geq 1 \rangle} \wedge \overset{@1}{\langle x_9 \leq 0 \rangle} \implies \langle x_6 \leq 1 \rangle$$



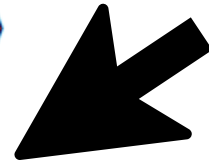
**Derive nogood
based on constraints**

Trail

$\langle x_9 \leq 0 \rangle$ @1
 $\langle x_1 \leq 2 \rangle$ @2
 $\langle x_2 \leq 3 \rangle$ @3

$c_1 \rightarrow \langle x_3 \geq 1 \rangle$
 $c_2 \rightarrow \langle x_4 \geq 1 \rangle$

$\langle x_6 \leq 1 \rangle$



**Backjump and propagate:
Decision level 3**

Conflict Analysis

$$\overset{@3}{\langle x_3 \geq 1 \rangle} \wedge \overset{@3}{\langle x_4 \geq 1 \rangle} \wedge \overset{@1}{\langle x_9 \leq 0 \rangle} \implies \langle x_6 \leq 1 \rangle$$

Trail

$$\langle x_9 \leq 0 \rangle$$

@1

$$\langle x_1 \leq 2 \rangle$$

@2

$$\langle x_2 \leq 3 \rangle$$

@3

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle x_6 \leq 1 \rangle$$

Backjump and propagate:
Decision level 3

Skipped
Decision level 4!

Conflict Analysis

$$\overset{@3}{\langle x_3 \geq 1 \rangle} \wedge \overset{@3}{\langle x_4 \geq 1 \rangle} \wedge \overset{@1}{\langle x_9 \leq 0 \rangle} \implies \langle x_6 \leq 1 \rangle$$



Trail

$$\langle x_9 \leq 0 \rangle$$

@1

$$\langle x_1 \leq 2 \rangle$$

@2

$$\langle x_2 \leq 3 \rangle$$

@3

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle y \geq 2 \rangle$$

@4

$$\langle x_5 \leq 2 \rangle$$

@5

$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

**Constraint c_6
Declares conflict**

Constraints

$$c_1 : 3x_2 + x_3 + x_9 \geq 10$$

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$$c_5 : 2x_8 - x_6 + x_9 \geq 0$$

$$c_6 : -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$x_i, y \in \{-10, 10\}$$

$$c_6: -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$\langle x_3 \overset{@3}{\geq} 1 \rangle \wedge \langle x_4 \overset{@3}{\geq} 1 \rangle \wedge \langle x_7 \overset{@5}{\geq} 2 \rangle \wedge \langle x_8 \overset{@5}{\geq} 1 \rangle \implies \perp$$

Conflict Nogood

$$c_6: -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_7 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_8 \geq 1 \rangle^{\textcircled{5}} \implies \perp$$


Decision level of
assignment

$$c_6: -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_7 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_8 \geq 1 \rangle^{\textcircled{5}} \implies \perp$$

Trail (Partial)

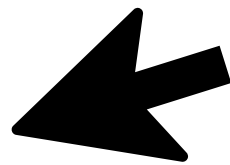
$$\langle x_5 \leq 2 \rangle^{\textcircled{5}}$$

$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

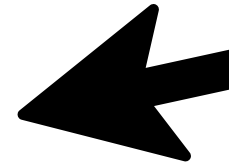


Last assigned
atomic constraint

$$c_6: -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$\langle x_3 \stackrel{@3}{\geq} 1 \rangle \wedge \langle x_4 \stackrel{@3}{\geq} 1 \rangle \wedge \langle x_7 \stackrel{@5}{\geq} 2 \rangle \wedge \boxed{\langle x_8 \stackrel{@5}{\geq} 1 \rangle} \implies \perp$$

$$c_5: 2x_8 - x_6 + x_9 \geq 0$$



$$\langle x_6 \stackrel{@5}{\geq} 2 \rangle \wedge \langle x_9 \stackrel{@1}{\leq} 0 \rangle \implies \boxed{\langle x_8 \stackrel{@5}{\geq} 1 \rangle}$$

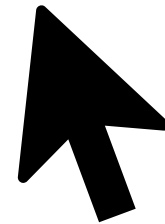
$$c_6: -x_3 - x_4 - x_7 - x_8 \geq -4$$

$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_7 \geq 2 \rangle^{\textcircled{5}} \wedge \boxed{\langle x_8 \geq 1 \rangle^{\textcircled{5}}} \implies \perp$$

$$c_5: 2x_8 - x_6 + x_9 \geq 0$$

Replace $\langle x_8 \geq 1 \rangle$
with its reason

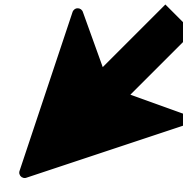
$$\langle x_6 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \boxed{\langle x_8 \geq 1 \rangle^{\textcircled{5}}}$$



$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_7 \geq 2 \rangle^{\textcircled{5}} \wedge \boxed{\langle x_8 \geq 1 \rangle^{\textcircled{5}}} \implies \perp$$

$$\langle x_6 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \boxed{\langle x_8 \geq 1 \rangle^{\textcircled{5}}}$$

New nogood!



$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_7 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_6 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \perp$$

$$\langle x_3 \stackrel{@3}{\geq} 1 \rangle \wedge \langle x_4 \stackrel{@3}{\geq} 1 \rangle \wedge \langle x_7 \stackrel{@5}{\geq} 2 \rangle \wedge \langle x_6 \stackrel{@5}{\geq} 2 \rangle \wedge \langle x_9 \stackrel{@1}{\leq} 0 \rangle \implies \perp$$

Trail (Partial)

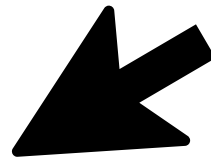
$$\langle x_5 \leq 2 \rangle \quad @5$$

$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$



Removed
dependency on
 $\langle x_8 \geq 1 \rangle$

$$\langle x_3 \stackrel{@3}{\geq} 1 \rangle \wedge \langle x_4 \stackrel{@3}{\geq} 1 \rangle \wedge \langle x_7 \stackrel{@5}{\geq} 2 \rangle \wedge \langle x_6 \stackrel{@5}{\geq} 2 \rangle \wedge \langle x_9 \stackrel{@1}{\leq} 0 \rangle \implies \perp$$

Trail (Partial)

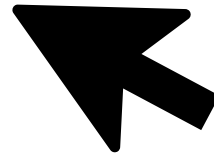
$$\langle x_5 \leq 2 \rangle \quad @5$$

$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$



Repeat procedure
from $\langle x_7 \geq 2 \rangle$

$$\langle x_3 \stackrel{@3}{\geq} 1 \rangle \wedge \langle x_4 \stackrel{@3}{\geq} 1 \rangle \wedge \boxed{\langle x_7 \stackrel{@5}{\geq} 2 \rangle} \wedge \langle x_6 \stackrel{@5}{\geq} 2 \rangle \wedge \langle x_9 \stackrel{@1}{\leq} 0 \rangle \implies \perp$$

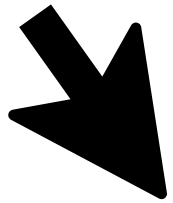
$$c_4: x_7 - x_6 \geq 0$$

$$\langle x_6 \stackrel{@5}{\geq} 2 \rangle \implies \boxed{\langle x_7 \stackrel{@5}{\geq} 2 \rangle}$$

$$\langle x_3^{\textcircled{3}} \geq 1 \rangle \wedge \langle x_4^{\textcircled{3}} \geq 1 \rangle \wedge \langle x_7^{\textcircled{5}} \geq 2 \rangle \wedge \langle x_6^{\textcircled{5}} \geq 2 \rangle \wedge \langle x_9^{\textcircled{1}} \leq 0 \rangle \implies \perp$$

$$c_4: x_7 - x_6 \geq 0$$

New nogood!



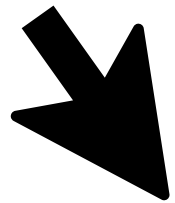
$$\langle x_6^{\textcircled{5}} \geq 2 \rangle \implies \langle x_7^{\textcircled{5}} \geq 2 \rangle$$

$$\langle x_3^{\textcircled{3}} \geq 1 \rangle \wedge \langle x_4^{\textcircled{3}} \geq 1 \rangle \wedge \langle x_6^{\textcircled{5}} \geq 2 \rangle \wedge \langle x_9^{\textcircled{1}} \leq 0 \rangle \implies \perp$$

$$\langle x_3^{\textcircled{3}} \geq 1 \rangle \wedge \langle x_4^{\textcircled{3}} \geq 1 \rangle \wedge \langle x_7^{\textcircled{5}} \geq 2 \rangle \wedge \langle x_6^{\textcircled{5}} \geq 2 \rangle \wedge \langle x_9^{\textcircled{1}} \leq 0 \rangle \implies \perp$$

$$c_4: x_7 - x_6 \geq 0$$

Propagates?



$$\langle x_6^{\textcircled{5}} \geq 2 \rangle \implies \langle x_7^{\textcircled{5}} \geq 2 \rangle$$

$$\langle x_3^{\textcircled{3}} \geq 1 \rangle \wedge \langle x_4^{\textcircled{3}} \geq 1 \rangle \wedge \langle x_6^{\textcircled{5}} \geq 2 \rangle \wedge \langle x_9^{\textcircled{1}} \leq 0 \rangle \implies \perp$$

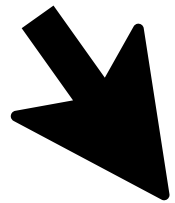
$$\langle x_3^{\textcircled{3}} \geq 1 \rangle \wedge \langle x_4^{\textcircled{3}} \geq 1 \rangle \wedge \langle x_7^{\textcircled{5}} \geq 2 \rangle \wedge \langle x_6^{\textcircled{5}} \geq 2 \rangle \wedge \langle x_9^{\textcircled{1}} \leq 0 \rangle \implies \perp$$

Propagates

at
decision
level 3!

“asserting” $c_4: x_7 - x_6 \geq 0$

$$\langle x_6^{\textcircled{5}} \geq 2 \rangle \implies \langle x_7^{\textcircled{5}} \geq 2 \rangle$$



$$\langle x_3^{\textcircled{3}} \geq 1 \rangle \wedge \langle x_4^{\textcircled{3}} \geq 1 \rangle \wedge \langle x_6^{\textcircled{5}} \geq 2 \rangle \wedge \langle x_9^{\textcircled{1}} \leq 0 \rangle \implies \perp$$

$$\langle x_3 \stackrel{@3}{\geq} 1 \rangle \wedge \langle x_4 \stackrel{@3}{\geq} 1 \rangle \wedge \langle x_6 \stackrel{@5}{\geq} 2 \rangle \wedge \langle x_9 \stackrel{@1}{\leq} 0 \rangle \implies \perp$$



$$\langle x_3 \stackrel{@3}{\geq} 1 \rangle \wedge \langle x_4 \stackrel{@3}{\geq} 1 \rangle \wedge \langle x_9 \stackrel{@1}{\leq} 0 \rangle \implies \langle x_6 \stackrel{@3}{\leq} 1 \rangle$$

Trail

$$\langle x_9 \leq 0 \rangle \quad @1$$

$$\langle x_1 \leq 2 \rangle \quad @2$$

$$\langle x_2 \leq 3 \rangle \quad @3$$

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle y \geq 2 \rangle \quad @4$$

$$\langle x_5 \leq 2 \rangle \quad @5$$

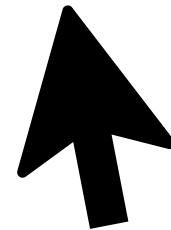
$$c_3 \rightarrow \langle x_6 \geq 2 \rangle$$

$$c_4 \rightarrow \langle x_7 \geq 2 \rangle$$

$$c_5 \rightarrow \langle x_8 \geq 1 \rangle$$

$$c_6 \rightarrow \perp$$

$$\overset{@3}{\langle x_3 \geq 1 \rangle} \wedge \overset{@3}{\langle x_4 \geq 1 \rangle} \wedge \overset{@1}{\langle x_9 \leq 0 \rangle} \implies \overset{@3}{\langle x_6 \leq 1 \rangle}$$



Trail

$$\langle x_9 \leq 0 \rangle \quad @1$$

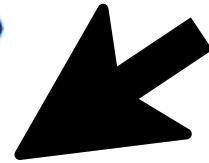
$$\langle x_1 \leq 2 \rangle \quad @2$$

$$\langle x_2 \leq 3 \rangle \quad @3$$

$$c_1 \rightarrow \langle x_3 \geq 1 \rangle$$

$$c_2 \rightarrow \langle x_4 \geq 1 \rangle$$

$$\langle x_6 \leq 1 \rangle$$



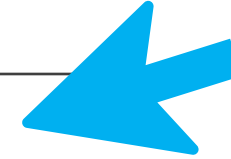
$$\overset{@3}{\langle x_3 \geq 1 \rangle} \wedge \overset{@3}{\langle x_4 \geq 1 \rangle} \wedge \overset{@1}{\langle x_9 \leq 0 \rangle} \implies \overset{@3}{\langle x_6 \leq 1 \rangle}$$

**Backjump and propagate:
Decision level 3**

Conflict Analysis

Algorithm 1 Conflict Analysis

Input: conflict explanation $(A_1 \wedge \dots \wedge A_n \implies \perp)$



Output: asserting nogood

$N \leftarrow$ conflict explanation

while N is not asserting **do**

 Let A be the most recently assigned atomic constraint in N

 Let $R \implies A$ be the explanation of A

 Replace A in N by R

end while

return N

Conflict Analysis

Algorithm 1 Conflict Analysis

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Output: asserting nogood

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while N is not asserting **do**

 Let A be the most recently assigned atomic constraint in N

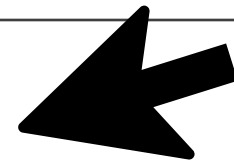
 Let $R \implies A$ be the explanation of A

 Replace A in N by R

end while

return N

Why does this work?



Why does conflict analysis work?

Propagation-based rewriting

Conflict preservation

Soundness

Termination

Why does conflict analysis work?

Propagation-based rewriting

“we only rewrite **propagated** atomic constraints”

Why does conflict analysis work?

Propagation-based rewriting

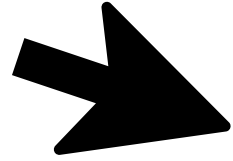
“we only rewrite **propagated** atomic constraints”

$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_7 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_8 \geq 1 \rangle^{\textcircled{5}} \implies \perp$$

$$\langle x_6 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \boxed{\langle x_8 \geq 1 \rangle^{\textcircled{5}}} \quad \leftarrow$$

$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_7 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_6 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \perp$$

Why does conflict analysis work?



Conflict preservation

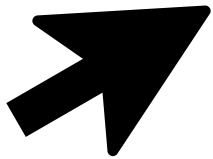
“each new nogood is also conflicting”

Why does conflict analysis work?

Conflict preservation

“each new nogood is also conflicting”

$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_7 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_8 \geq 1 \rangle^{\textcircled{5}} \implies \perp$$



$$\langle x_6 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \langle x_8 \geq 1 \rangle^{\textcircled{5}}$$

Started with
conflicting nogood

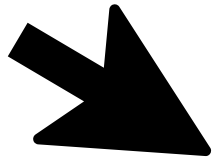
$$\langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_7 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_6 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \perp$$

Why does conflict analysis work?

Assigned “true (T)”

Conflict preservation

“each new nogood is also conflicting”



$$\overset{@3}{\langle x_3 \geq 1 \rangle} \wedge \overset{@3}{\langle x_4 \geq 1 \rangle} \wedge \overset{@5}{\langle x_7 \geq 2 \rangle} \wedge \overset{@5}{\langle x_8 \geq 1 \rangle} \implies \perp$$

$$\overset{@5}{\langle x_6 \geq 2 \rangle} \wedge \overset{@1}{\langle x_9 \leq 0 \rangle} \implies \overset{@5}{\langle x_8 \geq 1 \rangle}$$

$$\overset{@3}{\langle x_3 \geq 1 \rangle} \wedge \overset{@3}{\langle x_4 \geq 1 \rangle} \wedge \overset{@5}{\langle x_7 \geq 2 \rangle} \wedge \overset{@5}{\langle x_6 \geq 2 \rangle} \wedge \overset{@1}{\langle x_9 \leq 0 \rangle} \implies \perp$$

Why does conflict analysis work?

Conflict preservation

“each new nogood is also conflicting”

$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_7 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_8 \geq 1 \rangle^{\textcircled{5}} \implies \perp$$


$$\langle x_6 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \langle x_8 \geq 1 \rangle^{\textcircled{5}}$$

Must be “true” (T)

because
propagating

$$\langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_7 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_6 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \perp$$

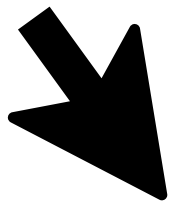
Why does conflict analysis work?

Conflict preservation

“each new nogood is also conflicting”

These must also
“true” (T)

$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_7 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_8 \geq 1 \rangle^{\textcircled{5}} \implies \perp$$



$$\langle x_6 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \langle x_8 \geq 1 \rangle^{\textcircled{5}}$$

$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_7 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_6 \geq 2 \rangle^{\textcircled{5}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \perp$$

Why does conflict analysis work?

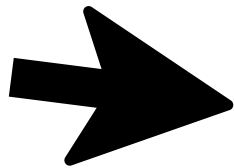
Soundness

“new nogoods are implied by the problem”

Why does conflict analysis work?

Soundness

“new nogoods are implied by the problem”



Rewriting is sound

Explanations are sound

Why does conflict analysis work?

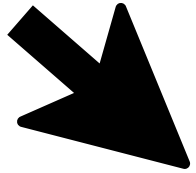
Termination

“eventually we derive an asserting nogood”

Why does conflict analysis work?

Termination

“eventually we derive an asserting nogood”



Atomic constraints are replaced based on recency

Finite trail size

Conflict Analysis

Algorithm 1 Conflict Analysis

Input: conflict explanation $(A_1 \wedge \dots \wedge A_n \implies \perp)$

Output: asserting nogood

$N \leftarrow$ conflict explanation

while N is not asserting **do**

 Let A be the most recently assigned atomic constraint in N

 Let $R \implies A$ be the explanation of A

 Replace A in N by R

end while

return $N = 0$

Propagation-based rewriting

Conflict preservation

Soundness

Termination

So far...

Conflict Analysis

Learn nogoods
by combining explanations

Potentially backjump

Up next...

Conflict-Driven Constraint Learning

Proofs of infeasibility/optimality

More general learning

Extended nogoods

Hypercube linears

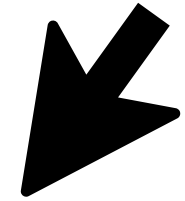
Search

Make a decision

Propagate based on constraints

If conflict, backtrack

Constraint Conflict-Driven ~~Clause~~ Learning (CDCL)



Make a decision **Conflict-driven branching
(VSIDS)**

Propagate based on constraints

If conflict, backtrack

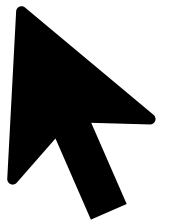
**“atomic constraints
appearing in conflicts
are important”**

Constraint Conflict-Driven ~~Clause~~ Learning (CDCL)

Make a decision **Conflict-driven branching
(VSIDS)**

Propagate based on constraints **...and learned nogoods**

If conflict, backtrack



Constraint Conflict-Driven ~~Clause~~ Learning (CDCL)

Make a decision **Conflict-driven branching
(VSIDS)**

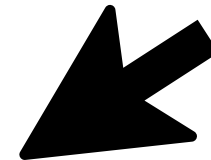
Propagate based on constraints **...and learned nogoods**

If conflict, ~~backtrack~~

Conflict analysis

Add learned nogood to database

Backjump



Constraint Conflict-Driven ~~Clause~~ Learning (CDCL)

Make a decision **Conflict-driven branching
(VSIDS)**

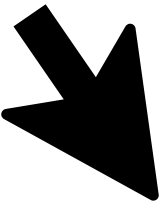
Propagate based on constraints **...and learned nogoods**

If conflict, ~~backtrack~~

Conflict analysis

Add learned nogood to database

Backjump



Restarts

**Nogood
Management**

Constraint Conflict-Driven ~~Clause~~ Learning (CDCL)

Make a decision **Conflict-driven branching
(VSIDS)**

Propagate based on constraints ...**and learned nogoods**

Restarts

If conflict, ~~backtrack~~

Conflict analysis

Add learned nogood to database

Backjump

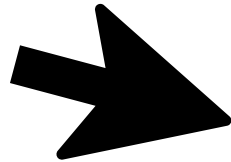
**Nogood
Management**

Constraint Conflict-Driven ~~Clause~~ Learning (CDCL)

CDCL stops when...

Constraint Conflict-Driven ~~Clause~~ Learning (CDCL)

CDCL stops when...

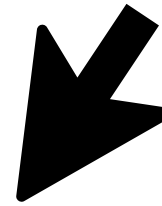


1. Feasible solution

Constraint Conflict-Driven ~~Clause~~ Learning (CDCL)

CDCL stops when...

1. Feasible solution



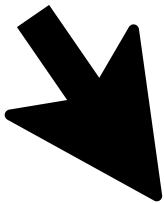
2. Learn infeasible nogood $T \rightarrow \perp$

Constraint Conflict-Driven ~~Clause~~ Learning (CDCL)

CDCL stops when...

1. Feasible solution

2. Learn infeasible nogood $\top \rightarrow \perp$



Solver trace is a certificate of infeasibility!

Proof-Theoretic View

$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

$$x \in \{0, 1, 2\}$$

$$y, z \in \{0, 1\}$$

Construct a certificate of infeasibility!

$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

$$x \in \{0, 1, 2\}$$

$$y, z \in \{0, 1\}$$

$$\langle x \geq 1 \rangle \wedge \langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$$



Claim

this nogood is implied by the constraints

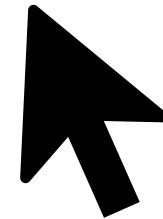
$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

$$x \in \{0, 1, 2\}$$

$$y, z \in \{0, 1\}$$

ID	Derived constraint		
c_3	c_1	$\models$	$\langle y \geq 0 \rangle \implies \langle x \leq 1 \rangle$
c_4	c_1	$\models$	$\langle x \geq 1 \rangle \implies \langle y \leq 0 \rangle$
c_5	c_2	$\models$	$\langle x \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \langle y \geq 1 \rangle$
c_6	$c_3 \wedge c_4 \wedge c_5$	$\models$	$\langle x \geq 1 \rangle \wedge \langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$



Proof of the claim

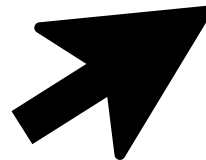
$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

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$$y, z \in \{0, 1\}$$

ID	Derived constraint	
c_3	$c_1 \models$	$\langle y \geq 0 \rangle \implies \langle x \leq 1 \rangle$
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c_5	$c_2 \models$	$\langle x \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \langle y \geq 1 \rangle$
c_6	$c_3 \wedge c_4 \wedge c_5 \models$	$\langle x \geq 1 \rangle \wedge \langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$



Explanation

$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

$$x \in \{0, 1, 2\}$$

$$y, z \in \{0, 1\}$$

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c_6	$c_3 \wedge c_4 \wedge c_5$	$\models$	$\langle x \geq 1 \rangle \wedge \langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$



**Derived through
conflict analysis**

$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

$$x \in \{0, 1, 2\}$$

$$y, z \in \{0, 1\}$$

$$\langle y \geq 0 \rangle \wedge \langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$$



New claim

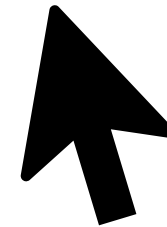
$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

$$x \in \{0, 1, 2\}$$

$$y, z \in \{0, 1\}$$

c_6	$c_3 \wedge c_4 \wedge c_5$	$\models$	$\langle x \geq 1 \rangle \wedge \langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$
c_7	c_6	$\models$	$\langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \langle x \leq 0 \rangle$
c_8	c_2	$\models$	$\langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \langle x \geq 1 \rangle$
c_9	$c_7 \wedge c_8$	$\models$	$\langle y \geq 0 \rangle \wedge \langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$



Proof of the claim

$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

$$x \in \{0, 1, 2\}$$

$$y, z \in \{0, 1\}$$

c_6	$c_3 \wedge c_4 \wedge c_5 \models \langle x \geq 1 \rangle \wedge \langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$
c_7	$c_6 \models \langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \langle x \leq 0 \rangle$
c_8	$c_2 \models \langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \langle x \geq 1 \rangle$
c_9	$c_7 \wedge c_8 \models \langle y \geq 0 \rangle \wedge \langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$

Previously proven claim used

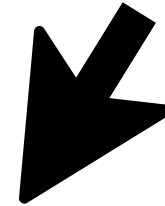
$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

$$x \in \{0, 1, 2\}$$

$$y, z \in \{0, 1\}$$

Refers to the bounds of variables



(not used so far)


$$\top \models \top \implies \langle y \geq 0 \rangle$$

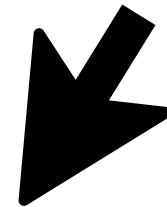
$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

$$x \in \{0, 1, 2\}$$

$$y, z \in \{0, 1\}$$

Final claim of infeasibility



$$\top \Rightarrow \perp$$

Proof of the final claim

$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

$$x \in \{0, 1, 2\}$$

$$y, z \in \{0, 1\}$$



c_9	$c_7 \wedge c_8$	$\models$	$\langle y \geq 0 \rangle \wedge \langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$
c_{10}	$\top$	$\models$	$\top \implies \langle y \geq 0 \rangle$
c_{11}	$\top$	$\models$	$\top \implies \langle y \leq 1 \rangle$
c_{12}	$\top$	$\models$	$\top \implies \langle z \leq 1 \rangle$
c_{13}	c_9	$\models$	$\langle y \geq 0 \rangle \wedge \langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$
c_{14}	$c_{10} \wedge c_{11} \wedge c_{12} \wedge c_{13}$	$\models$	$\top \implies \perp$

Certificate of infeasibility



$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

$$x \in \{0, 1, 2\}$$

$$y, z \in \{0, 1\}$$

ID	Derived constraint		
c_3	c_1	$\models$	$\langle y \geq 0 \rangle \implies \langle x \leq 1 \rangle$
c_4	c_1	$\models$	$\langle x \geq 1 \rangle \implies \langle y \leq 0 \rangle$
c_5	c_2	$\models$	$\langle x \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \langle y \geq 1 \rangle$
c_6	$c_3 \wedge c_4 \wedge c_5$	$\models$	$\langle x \geq 1 \rangle \wedge \langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$
c_7	c_6	$\models$	$\langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \langle x \leq 0 \rangle$
c_8	c_2	$\models$	$\langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \langle x \geq 1 \rangle$
c_9	$c_7 \wedge c_8$	$\models$	$\langle y \geq 0 \rangle \wedge \langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$
c_{10}	$\top$	$\models$	$\top \implies \langle y \geq 0 \rangle$
c_{11}	$\top$	$\models$	$\top \implies \langle y \leq 1 \rangle$
c_{12}	$\top$	$\models$	$\top \implies \langle z \leq 1 \rangle$
c_{13}	c_9	$\models$	$\langle y \geq 0 \rangle \wedge \langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$
c_{14}	$c_{10} \wedge c_{11} \wedge c_{12} \wedge c_{13}$	$\models$	$\top \implies \perp$

$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

$$x \in \{0, 1, 2\}$$

$$y, z \in \{0, 1\}$$

Explanations

+

Conflict Analysis

ID	Derived constraint		
c_3	c_1	$\models$	$\langle y \geq 0 \rangle \implies \langle x \leq 1 \rangle$
c_4	c_1	$\models$	$\langle x \geq 1 \rangle \implies \langle y \leq 0 \rangle$
c_5	c_2	$\models$	$\langle x \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \langle y \geq 1 \rangle$
c_6	$c_3 \wedge c_4 \wedge c_5$	$\models$	$\langle x \geq 1 \rangle \wedge \langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$
c_7	c_6	$\models$	$\langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \langle x \leq 0 \rangle$
c_8	c_2	$\models$	$\langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \langle x \geq 1 \rangle$
c_9	$c_7 \wedge c_8$	$\models$	$\langle y \geq 0 \rangle \wedge \langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$
c_{10}	$\top$	$\models$	$\top \implies \langle y \geq 0 \rangle$
c_{11}	$\top$	$\models$	$\top \implies \langle y \leq 1 \rangle$
c_{12}	$\top$	$\models$	$\top \implies \langle z \leq 1 \rangle$
c_{13}	c_9	$\models$	$\langle y \geq 0 \rangle \wedge \langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$
c_{14}	$c_{10} \wedge c_{11} \wedge c_{12} \wedge c_{13}$	$\models$	$\top \implies \perp$

$$c_1: 5x + 3y \leq 6$$

$$c_2: x + y + z \geq 3$$

$$x \in \{0, 1, 2\}$$

$$y, z \in \{0, 1\}$$

**Claim about
the instance,
not the solver!**

ID	Derived constraint		
c_3	c_1	$\models$	$\langle y \geq 0 \rangle \implies \langle x \leq 1 \rangle$
c_4	c_1	$\models$	$\langle x \geq 1 \rangle \implies \langle y \leq 0 \rangle$
c_5	c_2	$\models$	$\langle x \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \langle y \geq 1 \rangle$
c_6	$c_3 \wedge c_4 \wedge c_5$	$\models$	$\langle x \geq 1 \rangle \wedge \langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$
c_7	c_6	$\models$	$\langle y \geq 0 \rangle \wedge \langle z \leq 1 \rangle \implies \langle x \leq 0 \rangle$
c_8	c_2	$\models$	$\langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \langle x \geq 1 \rangle$
c_9	$c_7 \wedge c_8$	$\models$	$\langle y \geq 0 \rangle \wedge \langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$
c_{10}	$\top$	$\models$	$\top \implies \langle y \geq 0 \rangle$
c_{11}	$\top$	$\models$	$\top \implies \langle y \leq 1 \rangle$
c_{12}	$\top$	$\models$	$\top \implies \langle z \leq 1 \rangle$
c_{13}	c_9	$\models$	$\langle y > 0 \rangle \wedge \langle y \leq 1 \rangle \wedge \langle z \leq 1 \rangle \implies \perp$
c_{14}	$c_{10} \wedge c_{11} \wedge c_{12} \wedge c_{13}$	$\models$	$\top \implies \perp$

(See our CP'26 paper)

Stronger Conflict Analysis

Nogood Propagation

$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \langle x_6 \leq 1 \rangle$$

Atomic constraint

$\{?, \textcolor{blue}{\top}, \textcolor{brown}{\perp}\}$

Nogood Propagation

$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \langle x_6 \leq 1 \rangle$$

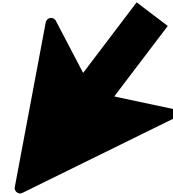
Atomic constraint

$\{?, \textcolor{blue}{T}, \textcolor{brown}{\perp}\}$

$x_3 \in [6, 10]$

$\langle x_3 \geq 5 \rangle$

$\textcolor{blue}{T}$

True


Nogood Propagation

$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \langle x_6 \leq 1 \rangle$$

Atomic constraint

$\{?, \textcolor{blue}{\top}, \textcolor{brown}{\perp}\}$

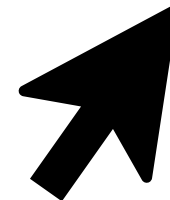
$x_3 \in [6, 10]$

$\langle x_3 \geq 5 \rangle$ $\textcolor{blue}{\top}$

$x_3 \in [6, 10]$

$\langle x_3 \leq 5 \rangle$ $\textcolor{brown}{\perp}$

False



Nogood Propagation

$$\langle x_3 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_4 \geq 1 \rangle^{\textcircled{3}} \wedge \langle x_9 \leq 0 \rangle^{\textcircled{1}} \implies \langle x_6 \leq 1 \rangle$$

Atomic constraint

$\{?, \top, \perp\}$

$$x_3 \in [6, 10]$$

$$\langle x_3 \geq 5 \rangle$$

$\top$ Unassigned

$$x_3 \in [6, 10]$$

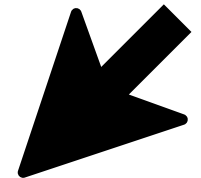
$$\langle x_3 \leq 5 \rangle$$

$\perp$

$$x_3 \in [0, 10]$$

$$\langle x_3 \geq 5 \rangle$$

?



Nogood Propagation

$$A \wedge B \wedge C \wedge D \Rightarrow \perp$$

$$A, B, C, D \in \{?, \textcolor{blue}{T}, \textcolor{orange}{\perp}\}$$

No propagation when all unassigned



$$A \wedge B \wedge C \wedge D \Rightarrow \perp$$

$$A = ?$$

$$B = ?$$

$$C = ?$$

$$D = ?$$

No propagation...



$$\textcolor{blue}{A} \wedge B \wedge C \wedge D \Rightarrow \perp$$

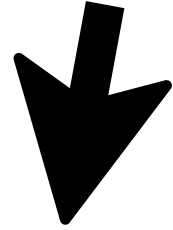
$$\textcolor{blue}{A} = \top$$

$$B = ?$$

$$C = ?$$

$$D = ?$$

Still no propagation...



$$\textcolor{blue}{A} \wedge \textcolor{blue}{B} \wedge \textbf{\textit{C}} \wedge \textbf{\textit{D}} \Rightarrow \perp$$

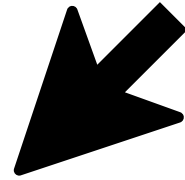
$$\textcolor{blue}{A} = \top$$

$$\textcolor{blue}{B} = \top$$

$$\textbf{\textit{C}} = ?$$

$$\textbf{\textit{D}} = ?$$

Unit Propagation!



$$\textcolor{blue}{A} \wedge \textcolor{blue}{B} \wedge \textcolor{blue}{C} \wedge \textbf{D} \Rightarrow \perp$$

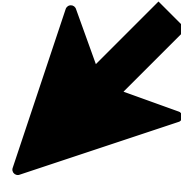
$$\textcolor{blue}{A} = \top$$

$$\textcolor{blue}{B} = \top$$

$$\textcolor{blue}{C} = \top$$

$$\textbf{D} = ?$$

D forced to $\perp$



$$A \wedge B \wedge C \wedge D \Rightarrow \perp$$

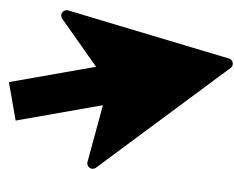
$$A = \top$$

$$B = \top$$

$$C = \top$$

$$D = \perp$$

$$A \wedge B \wedge C \wedge D \Rightarrow \perp$$



$$A \wedge B \wedge C \Rightarrow \bar{D}$$

$$A = \top$$

$$B = \top$$

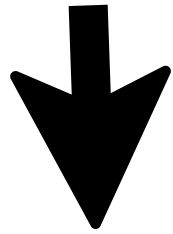
$$C = \top$$

$$D = \perp$$

Nogood Propagation

$$A_1 \wedge A_2 \wedge \cdots \wedge A_n \Rightarrow \perp$$

Nogood Propagation



$$A_1 \wedge A_2 \wedge \cdots \wedge A_n \Rightarrow \perp$$

Conflict

$$A_i = \top$$

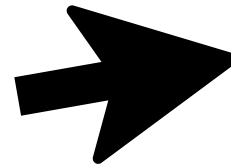
Nogood Propagation

$$A_1 \wedge A_2 \wedge \cdots \wedge A_n \Rightarrow \perp$$

Conflict

$$A_i = \top$$

“unit propagation”



Propagation

$$A_{1\dots(n-1)} = \top$$

$$A_n = \perp$$

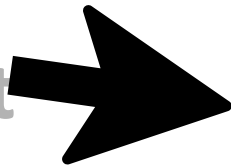
$$A_1 \wedge A_2 \wedge \cdots \wedge A_{n-1} \Rightarrow \overline{A_n}$$

Nogood Propagation

$$A_1 \wedge A_2 \wedge \cdots \wedge A_n \Rightarrow \perp$$

Conflict

$$A_i = \top$$



**Can we do
more
propagation?**

Propagation

$$A_{1\dots(n-1)} = \top$$

$$A_n = \perp$$

$$A_1 \wedge A_2 \wedge \cdots \wedge A_{n-1} \Rightarrow \overline{A_n}$$

Extended Propagation



$$A \wedge B \wedge C \wedge D \Rightarrow \perp$$

$$A = ?$$

$$B = ?$$

$$C = ?$$

$$D = ?$$

No propagation...



$$\textcolor{blue}{A} \wedge B \wedge C \wedge D \Rightarrow \perp$$

$$\textcolor{blue}{A} = \top$$

$$B = ?$$

$$C = ?$$

$$D = ?$$

Still no propagation according to unit propagation...



$$\textcolor{blue}{A} \wedge \textcolor{blue}{B} \wedge \textbf{C} \wedge \textbf{D} \Rightarrow \perp$$

$$\textcolor{blue}{A} = \top$$

$$\textcolor{blue}{B} = \top$$

$$\textbf{C} = ?$$

$$\textbf{D} = ?$$

...but if we look at the atomic constraints...



$$\langle x \geq 5 \rangle \wedge \langle y \geq 2 \rangle \wedge \langle z \geq 1 \rangle \wedge \langle z \leq 3 \rangle \Rightarrow \perp$$

$$\langle x \geq 5 \rangle = \top$$

$$\langle y \geq 3 \rangle = \top$$

$$\langle z \geq 1 \rangle = ?$$

$$\langle z \leq 3 \rangle = ?$$

$$\langle x \geq 5 \rangle \wedge \langle y \geq 2 \rangle \wedge \langle z \geq 1 \rangle \wedge \langle z \leq 3 \rangle \Rightarrow \perp$$

$$\langle x \geq 5 \rangle \wedge \langle y \geq 2 \rangle \Rightarrow \langle z \neq 1 \rangle \wedge \langle z \neq 2 \rangle \wedge \langle z \neq 3 \rangle$$

$$\langle x \geq 5 \rangle = \top$$

$$\langle y \geq 3 \rangle = \top$$

$$\langle z \geq 1 \rangle = ?$$

$$\langle z \leq 3 \rangle = ?$$



We can propagate!

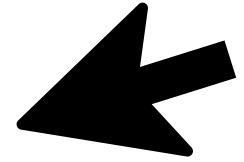
**“extended
nogood propagation”**

Conflict Analysis with Extended Nogood Propagation

Algorithm 1 Conflict Analysis

Input: conflict explanation $(A_1 \wedge \dots \wedge A_n \implies \perp)$

Output: ~~asserting nogood~~ propagating nogood



$N \leftarrow$ conflict explanation

while ~~N is not asserting~~ **do** N is not propagating

 Let A be the most recently assigned atomic constraint in N

 Let $R \implies A$ be the explanation of A

 Replace A in N by R

end while

return N

Conflict Analysis with Extended Nogood Propagation

Algorithm 1 Conflict Analysis

Input: conflict explanation $(A_1 \wedge \dots \wedge A_n \implies \perp)$

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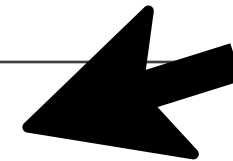
 Replace A in N by R

end while

return N

(see our CP'26 paper...)

More general learning!



Conflict analysis
is about
combining constraints!

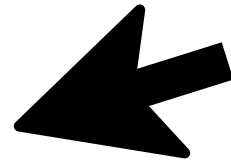
Problem with two constraints

$$\sum w_i x_i \geq k \quad \wedge \quad \sum w_i x_i \leq k - 1$$

Problem with two constraints

$$\sum w_i x_i \geq k \quad \wedge \quad \sum w_i x_i \leq k - 1$$

Infeasible!

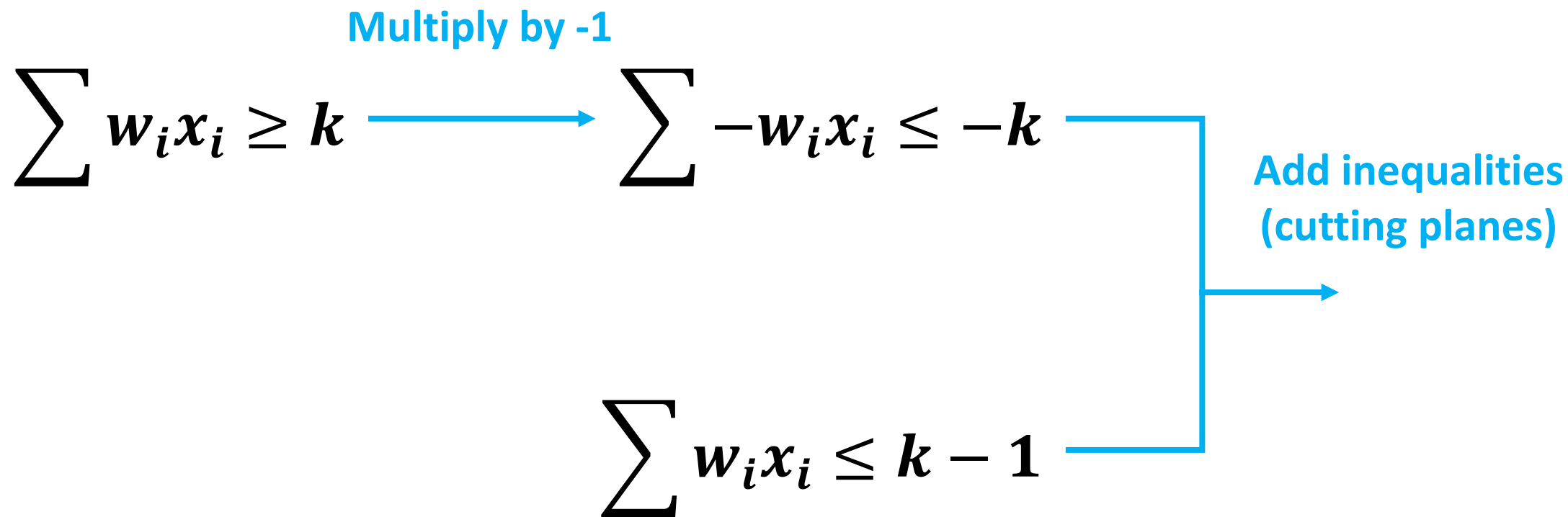


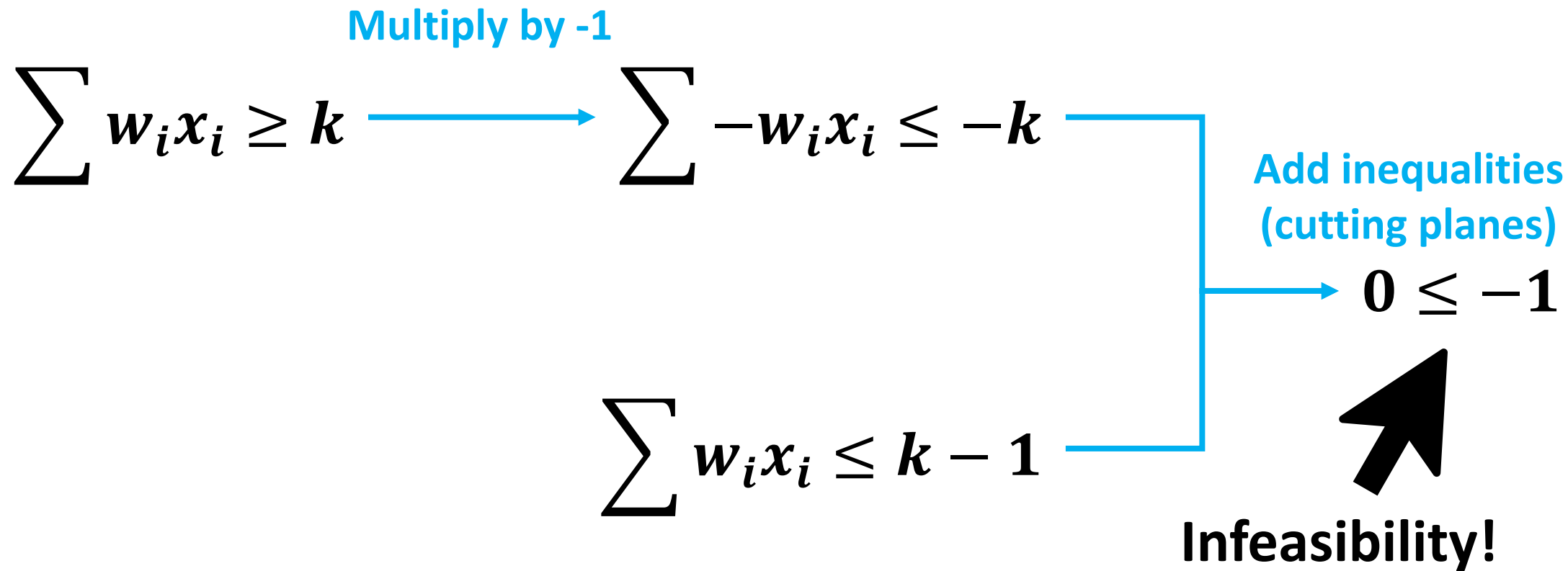
Easy problem, but exponential time for
our search with conflict analysis!

Cutting planes: a better way...

$$\sum w_i x_i \geq k \xrightarrow{\text{Multiply by -1}} \sum -w_i x_i \leq -k$$

$$\sum w_i x_i \leq k - 1$$





Exponentially better
(on this example)

Combining Constraints

1. “Rewrite” nogoods

two nogoods $\rightarrow$ new nogood



Complete

Combining Constraints

1. “Rewrite” nogoods

two nogoods $\rightarrow$ new nogood

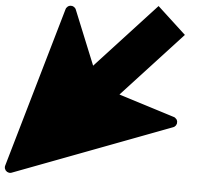
Complete

2. Cutting planes

two inequalities $\rightarrow$ new inequality

Incomplete

(complete only in
real-valued space)



Combining Constraints

1. “Rewrite” nogoods

two nogoods $\rightarrow$ new nogood

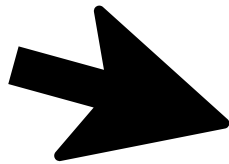
Complete

2. Cutting planes

two inequalities $\rightarrow$ new inequality

Incomplete

(complete only in
real-valued space)



Can we combine the two?

Hypercube Linear Constraints

$$A_1 \wedge A_2 \wedge \cdots \wedge A_n \Rightarrow \sum w_i x_i \geq k$$

Hypercube Linear Constraints

$$A_1 \wedge A_2 \wedge \cdots \wedge A_n \Rightarrow \sum w_i x_i \geq k$$

Generalises
nogoods

$$A_1 \wedge A_2 \wedge \cdots \wedge A_n \Rightarrow \perp$$

Generalises
linear inequalities

$$\top \Rightarrow \sum w_i x_i \geq k$$

Conflict Analysis with Hypercube Linears

Algorithm 1 Conflict Analysis

Conflict hypercube linear

Input: ~~conflict explanation $(A_1 \wedge \dots \wedge A_n \implies \perp)$~~

Output: ~~asserting nogood~~ propagating hypercube linear

$N \leftarrow$ conflict explanation

while ~~N is not asserting~~ **do** N is not propagating

Let A be the most recently assigned atomic constraint in N

~~Let $R \implies A$ be the explanation of A~~ Hypercube linear HL

~~Replace A in N by R~~ that propagated A

end while

return N

$N \leftarrow$ Combine HL and N

Conflict Analysis with Hypercube Linears

Algorithm 1 Conflict Analysis

Conflict hypercube linear

Input: ~~conflict explanation $(A_1 \wedge \dots \wedge A_n \implies \perp)$~~

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~~Let $R \implies A$ be the explanation of A~~ Hypercube linear HL

~~Replace A in N by R~~

that propagated A

end while

return N

$N \leftarrow$ Combine HL and N



Analogous as before!

Conflict Analysis with Hypercube Linears

Algorithm 1 Conflict Analysis

Conflict hypercube linear

Input: ~~conflict explanation $(A_1 \wedge \dots \wedge A_n \Rightarrow \perp)$~~

Output: ~~asserting nogood~~ propagating hypercube linear

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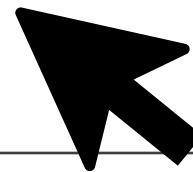
~~Let $R \Rightarrow A$ be the explanation of A~~ Hypercube linear HL

~~Replace A in N by R~~ that propagated A

end while

return N

$N \leftarrow$ Combine HL and N



More complex!
(see our CPAIOR'26 paper...)

Part 1

Propagators and Explanations

Linear inequality: $\sum w_i x_i \geq c$

Max: $x = \max(y, z)$

All-Different

Checkers

Part 2

Conflict Analysis

Combine constraints

Proofs

Beyond LCG

Extended nogoods

Hypercube linears